Psychological Review

“Associative learning is sufficient to explain infants’ action understanding”

--Manuscript Draft--

Manuscript Number:	REV-2025-1866
Full Title:	“Associative learning is sufficient to explain infants’ action understanding”
Abstract:	One of the many challenges that young learners face is learning how to navigate the world. A key prerequisite for such navigation is distinguishing between efficient and inefficient actions. Several domain-specific proposals have been offered to explain how infants achieve this. The current paper departs from these proposals by arguing that infants' behaviors in studies that evaluate their action understanding can be explained by a combination of real-world experience with different kinds of actions and real-time, online domain-general associative learning. I demonstrate that when this proposal is implemented in a connectionist artificial neural network model, the model (and by extension, the proposal that it implements) can account for a range of findings from studies that use some version of the original Gergely et al. (1995) paradigm. The results of four simulation studies indicate that associative learning alone can explain several aspects of infants’ looking behavior in these experiments. Specifically, the model successfully captures the findings that infants look longer at inefficient actions than efficient ones, respond similarly to animate and inanimate agents, infer goals from observed constraints, and represent efficiency continuously. These results challenge the necessity of domain-specific accounts such as the teleological stance or naïve utility calculus. Instead, they suggest associative learning alone can provide a more parsimonious explanation of infants’ action understanding.
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07/17/2025

Dear Dr. Elena L. Grigorenko,

I have submitted a manuscript titled, "Associative learning is sufficient to explain infants' action understanding" for consideration as a theoretical note at *Psychological Review*.

This manuscript introduces a novel theoretical account of early infant action understanding. Prior work has argued that infants interpret actions—such as an agent moving around a barrier to reach a goal object—using abstract, theory-like knowledge supported by domain-specific mechanisms (e.g., a teleological stance [Gergely & Csibra, 2003] or a naïve utility calculus [Jara-Ettinger et al., 2020]). Here, I propose an alternative view based on the idea that infants' behavior in such tasks can be explained by domain-general associative learning mechanisms applied to low-level perceptual features, in conjunction with real-world experience observing different types of actions (e.g., curved vs. linear motion).

To evaluate this claim, I implement the proposed account in a connectionist neural network model. Across 4 simulations, the model reproduces key findings from the infant action understanding literature. These include infants' preference for efficient over inefficient actions, insensitivity to animacy cues when interpreting efficiency, ability to infer goals or constraints from partial information, and graded sensitivity to relative action cost. These results suggest that domain-general associative processes, coupled with prior experience, may be sufficient to explain infants' behavior, and that conceptually rich or innately specified mechanisms may not be necessary.

More broadly, the manuscript contributes to ongoing debates in the developmental cognitive sciences regarding the origins of knowledge by offering a mechanistic, unified, and parsimonious alternative to domain-specific accounts. This manuscript will be of interest to readers concerned with developmental theory, cognitive mechanisms, and the role of computational modeling for testing cognitive theories.

This manuscript is original, has not been published, and is not under review elsewhere. I have no conflicts of interest to disclose. All simulation code and data are available, and the link to the OSF repository is in the manuscript.

Dr. Michael Waldmann or Dr. Hongjing Lu would be suitable action editors.

Recommended reviewers include the following individuals, all of whom are content experts with relevant expertise in developmental science, learning mechanisms, or computational modeling. I have no conflicts of interest with any of them:

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Thank you for considering this submission. I look forward to your response.

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Associative learning is sufficient to explain infants' action understanding

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Data availability statement: The simulation data for these simulations can be found here https://osf.io/9eqpf/?view_only=62f752417a9141c7b0509a998039ad13.

Conflicts of interests: none

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"Associative learning is sufficient to explain infants' action understanding"

Abstract

One of the many challenges that young learners face is learning how to navigate the world. A key prerequisite for such navigation is distinguishing between efficient and inefficient actions. Several domain-specific proposals have been offered to explain how infants achieve this. The current paper departs from these proposals by arguing that infants' behaviors in studies that evaluate their action understanding can be explained by a combination of real-world experience with different kinds of actions and real-time, online domain-general associative learning. I demonstrate that when this proposal is implemented in a connectionist artificial neural network model, the model (and by extension, the proposal that it implements) can account for a range of findings from studies that use some version of the original Gergely et al. (1995) paradigm. The results of four simulation studies indicate that associative learning alone can explain several aspects of infants' looking behavior in these experiments. Specifically, the model successfully captures the findings that infants look longer at inefficient actions than efficient ones, respond similarly to animate and inanimate agents, infer goals from observed constraints, and represent efficiency continuously. These results challenge the necessity of domain-specific accounts for action understanding. Instead, they suggest associative learning and real-world, third-person action experience provides a more parsimonious explanation of infants' action understanding.

"Associative learning is sufficient to explain infants' action understanding"

A key task for young learners is discovering how to navigate the world. One prerequisite for successful navigation is distinguishing between efficient and inefficient actions. For instance, a child will learn that it is easier and more efficient to consume smaller bites of food than to attempt to fit entire handfuls of food into their mouth. Likewise, a child will learn that using stairs to reach the top of a slide is safer and more efficient than attempting clumsily to climb it from the bottom. Finally, a child will discover that opening the gate to their play area by grabbing and pulling its handle is physically less taxing, safer, and more efficient than attempting to clamber over the gate. Although young learners may readily discover what makes some actions more efficient than others, the wide range of contexts they encounter poses significant challenges for acquiring this knowledge more generally.

Yet, despite its apparent difficulty, recognizing that some actions are more efficient than others is ostensibly in place in infancy (Csibra et al., 1999; Gergely et al., 1995). Evidence for this comes from studies that have used the violation-of-expectation (VoE) paradigm (e.g., Baillargeon et al., 1995). In this method, infants are habituated or familiarized either to an event or set of events designed to establish an expectation of some kind or to acquaint them with the set up of the experiment. In a subsequent test phase, infants are shown events that either conform to or violate the expectations that an adult observer might have formed during the initial phase (e.g., Stahl & Kibbe, 2022). A number of important insights about infants' early number concepts (e.g. McCrink & Wynn, 2004; Wynn, 1992), physical concepts (e.g., Kim & Spelke, 1992; Spelke et al., 1992), moral concepts (e.g., Geraci & Surian, 2011; Hamlin et al., 2007; Schmidt & Sommerville, 2011; Woo et al., 2017), object concepts (e.g., Baillargeon, 1987; Baillargeon et al., 1985; Kellman & Spelke, 1983; Needham & Baillargeon, 1997), and causal concepts (e.g.,

Kotovsky & Baillargeon, 1998; Leslie & Keeble, 1987; Muentener & Carey, 2010) have emerged from research using this method. The general takeaway from this research is that infants possess sophisticated capacities that support learning in each of these domains from early in development.

Two of the first studies to use the VoE method to assess infant action understanding in particular was that by Gergely et al. (1995) and Csibra et al. (1999). These authors habituated infants between 6 and 12 months of age to an event in which an animated ball jumped over a barrier to reach another animated ball. Infants were then shown two barrierless test events. In the *old action* test event, the small circle followed the familiar curved motion path to reach the large circle. In contrast, in the *new action* test event, the small circle moved along a straight path to reach the large circle. The results of these studies revealed that infants 9 months of age and older looked longer at the old action event than at the new action event. A control condition testing whether infants looked longer at the old action than to the new action event because of an a priori preference for curved trajectories found that infants looked equally long at both events when the small circle jumped beside rather than over the barrier.

This finding has since been extended in important ways. For example, it is now known that infants' expectation for efficient actions applies not only to people (e.g., Phillips & Wellman, 2005; Sodian et al., 2004; Woodward & Sommerville, 2000) and animals (e.g., Schwier et al., 2006) but also to geometric shapes (Csibra et al., 2003; Gergely et al., 1995) and inanimate objects such as robots (Kamewari et al., 2005). Additionally, although infants fail to distinguish inefficient and efficient actions if the target or goal object is absent (e.g., Phillips & Wellman, 2005), they recognize that two actions can be inefficient (or efficient) even if one is more of one kind than the other (Liu & Spelke, 2017). Beyond infants' processing of others' actions, research

shows that how infants imitate another person's actions depends on the constraints imposed on those actions. For example, 14-month-olds use their foreheads to activate a light if they observe a hands-free agent do the same thing but use their hands to activate the same light if the agent's hands are bound (e.g., Gergely et al., 2002).

Rich proposals for the origins of infant action understanding

Several proposals have been put forward to explain the mechanistic basis of infant action understanding. These proposals fall roughly into two classes. The first is "rich" proposals for the origins of infant action understanding. According to these accounts, infants possess from birth or shortly thereafter domain-specific mechanisms, or mechanisms that are triggered by a single input type and operate in a single domain, that support abstract and conceptually rich knowledge about how the various objects and entities in the world behave and act. For example, some argue that infants' knowledge about how animate entities act is subserved by a *core knowledge system* for agents, whereas their knowledge about inanimate objects is supported by a core system for objects (e.g., Carey, 2009; Spelke, 2022). The core system for agents enables infants to know that agents are goal-directed, self-propelled, can move on irregular trajectories, and can cause other agents to act at a distance. In contrast, the core system for objects allows infants to know that objects do not move in the absence of physical contact from other objects or agents, are bounded and cohesive, and tend to move on linear trajectories.

Conceptually similar proposals have been put forward by other theorists. For example, Leslie (1995) argued that infants' knowledge about agents and objects is supported by a *theory of body* and a *theory of mind* module. These modules enable infants to know that animate entities, but not inanimate objects, act in accordance with their beliefs, desires, and goals and can move in

the absence of external contact from other agents or objects via an internal and renewable source of energy. Similarly, Gelman (1990) argued that infants' knowledge about objects and agents is subserved by innate *skeletal causal principles*. The function of these principles is to direct infants' and young children's attention toward perceptual cues that distinguish agents from objects. For example, in the presence of an agent, children may focus on its legs, eyes, or skin because these cues may indicate that the agent is capable of self-generated motion. In contrast, in the presence of an inanimate object, children may attend to its external parts (e.g., wheels), its rigid structure, or its mechanical components because these features may indicate that an external force is needed for movement.

In addition to these broad accounts of how agents and objects move and behave, specific proposals have been developed to explain infants' looking behaviors in studies like that by Gergely et al. (1995). One is the *teleological stance* (Csibra & Gergely, 1998; Gergely & Csibra, 2003). Proponents of this account maintain that if an agent's or object's behavior satisfies the "rationality principle" such that it is seen as the most efficient and rational way to bring about an end state given the physical context, then the end states will be seen as the "goal", the agent's behaviors as a "means" to the goal, and the physical context as a set of "constraints". In other words, an agent's actions are seen as goal-directed only if they are the most efficient actions given the environmental constraints. Thus, the reason infants in the Experimental conditions of the studies mentioned above looked longer during a test phase when a first agent followed a curved path to reach a second object than when it follows a straight path to reach the same object is because this action is no longer justified when the barrier that was present during habituation is absent. In contrast, the reason infants in the Control condition looked equally long at the curved and straight paths is that they had not formed an efficiency-based expectation during habituation,

given that the agent's actions during habituation were not rational given the environmental constraints. Crucially, this account is "non-mentalistic" in that infants can recognize an action as goal-directed without considering or inferring an agent's mental states, such as its desires, beliefs, or intentions; instead, infants apply a "reality-based" interpretation of an action in terms of the alignment between that action, an end state, and external environmental constraints.

A conceptually related account is the *naïve utility calculus* (e.g., Jara-Ettinger et al., 2016; Jara-Ettinger et al., 2020). This proposal concerns how agents choose actions and how observers process those actions. On the agent side, actors evaluate possible plans—that is, sequences of actions that lead to a goal—and select the plan that maximizes expected subjective utilities. A subjective utility is the difference between the rewards that result from an outcome and the cost of executing the plan. Because making this difference as large as possible depends on minimizing unnecessary costs, agents are expected to carry out plans that are as efficient as possible, given the constraints of the environment. These utilities are considered "subjective" because the perceived rewards of an outcome and the perceived costs of a plan may differ across agents. On the observer side, observers must infer the rewards and costs that drove an agent's behavior from the behavior itself. The way observers solve this computational problem is via Bayesian inference. In other words, learners can estimate the probability of the reward of an outcome and the cost of executing a plan to achieve that outcome given different actions—that is, $p(\mathrm{C}, \mathrm{R}|\mathrm{A})$—by computing the product of two terms. One is the likelihood of an action given a reward and associated cost—that is, $p(A|C, R)$. The other is the likelihood of the reward and cost by itself—that is, $p(C, R)$.

This account can also explain infants' looking behavior in studies like Gergely et al. (1995). Consider the Experimental condition, for example. In that condition, the agent engages in

a costly action to reach a goal (i.e., jumping over a barrier). Because observers assume that agents are strictly utility maximizers, the observer (in this case, the infant) reasons that this action only makes sense if the reward for reaching the goal is large enough to outweigh the cost of jumping over the barrier. However, this same action no longer makes sense once the barrier is removed: The agent could have taken a path that was shorter, more efficient, and less effortful. The fact that both paths lead to the same size reward despite one being more costly (the jumping path) than the other (the straight path) implies that the straight path has greater utility; that is, the difference between the reward and the cost is larger for the straight path than the curved path. As such, the agent should choose to execute this plan. A corollary of this is that infants will look longer at the jumping path test event than at the straight path test event because their expectation for efficient, utility-maximizing behavior is violated in the former but not in the latter. In contrast, in the control condition, infants never formed a clear expectation about which path is more efficient. This is because the barrier was not included in the habituation event. As a result, they look equally at the test events. In this way, NUC explains both patterns of looking by assuming that infants prefer actions that make sense in terms of getting the most reward for the least cost.

Although each of the perspectives presented in this section differ in their specifics, what is common among them is the notion that infants possess abstract, conceptually rich knowledge and can make theory-like inferences about how agents and objects act and behave in the world, and that this knowledge and these inferences are supported by domain-specific or specialized processes that are operational and possibly innately endowed.

Lean proposals for the origins of infant action understanding

The second class is “lean” proposals of action understanding (e.g., Cohen et al., 2002; Colunga & Smith, 2005; Cohen & Oakes, 2003; Jones & Smith, 1993; Quinn & Eimas, 1997; Rakison & Lupyan, 2008; Sloutsky, 2010). The crux of these proposals is that infants’ behavior in VoE studies—of which the study by Gergely et al. (1995) is a special case—do not reflect specialized processes for reasoning about agent and object action. Instead, infants’ behavior in these studies can be explained by the operation of domain-general learning mechanisms (e.g., associative learning, habituation, conditioning, pattern learning, statistical learning, etc.) that operate over low-level, surface-based perceptual features and information. Domain-general mechanisms are sometimes considered “all purpose” because they operate on many kinds of input and support learning across a wide range of domains. Although the theories that constitute this class of perspectives differ in their specifics, each places considerable importance on the surface properties of objects and each emphasizes the idea that infants’ concepts for animates and inanimates are formed much more slowly than the available data may suggest. An implication of this account is that infants’ behaviors in VoE studies do not reflect, sophisticated adult-like abilities, conceptually rich knowledge, or abstract, theory-like inferences; rather, such abilities and knowledge emerge gradually over time through a slow, effortful, and piecemeal process that begins in early infancy and continues through childhood to adulthood.

To see how lean accounts explain infants’ looking behaviors in VoE studies, it can help to consider how they explain specific findings in the developmental cognitive science literature. To that end, consider the finding that infants ostensibly possess a theory of mind and understand that people can have beliefs that are incongruous with reality (i.e., false beliefs). In one of the first studies on this topic in infants, Onishi and Baillargeon (2005) showed infants a stage that contained a toy watermelon slice positioned between a yellow box and a green box. A previously

unseen actor then emerged from behind a wall, picked up the watermelon slice, and placed it into one of the boxes (e.g., the green box). The actor's visual access to the contents on the stage was then removed for several seconds, after which the actor returned and reached into the green box (as if to retrieve the toy). These trials established that the actor desired the toy. The experimenter then manipulated whether the actor held a true or false belief. In the true-belief condition, the actor watched as the toy was moved from the green box to the yellow box. In the false-belief condition, the actor again watched the toy move from the green box to the yellow box. The actor then moved from view behind the back wall, at which point the toy was moved back to the green box. Thus, the actor held a false belief that the toy was in the yellow box. The authors found that infants looked longer when the actor reached in the green box than when it reached in the yellow box if it held a true belief about the toy's location. Infants showed this same pattern of looking when the actor held a false belief. Onishi and Baillargeon (2005) interpreted these results to mean that infants understand that people act in accordance with their beliefs and that they may possess beliefs that are inconsistent with reality.

Perner and Ruffman (2005) challenged this interpretation by arguing that infants' responses to the test events may be based on learned associations among low-level aspects of the events rather than on an understanding of others' mental states. These authors specifically argued that infants' looking behaviors during the test events reflected tripartite associations that were formed during the familiarization and belief-induction phases. For instance, seeing the toy move from the green box to the yellow box in the presence of the actor in both belief conditions could have led infants to form a simple actor-toy-yellow-box association. A consequence of such an association is that events that are consistent with it—such as the test event in which the actor retrieves the toy from the yellow box—should lead to little (or no) increase in looking, whereas

events that are inconsistent with the established association—such as the test events in which the actor retrieves the toy from the green box—should lead to an increase in looking. As this alternative lean explanation makes clear, infants' looking behaviors in Onishi and Baillargeon's (2005) study can be explained by simple associative learning without recourse to an early emerging understanding of others' mental states.

Similar lean proposals have been used to explain infants' looking behaviors in other VoE studies. For example, it has been suggested that infants possess innately a domain-specific "accumulator" mechanism that enables them precisely to add and subtract small and large numbers (McCrink & Wynn, 2004; Wynn, 1992, 1995). One challenge to this view is that infants' looking patterns in these studies reflect imprecise quantity or amount estimation rather than exact numerical reasoning (e.g., Cohen & Marks, 2002). In other words, infants may simply expect addition to result in more of something and subtraction to result in less of something, even if the precise quantities that are shown are inconsistent with the correct outcome. Similarly, some argue that infants possess an "innate moral core" that, among other things, enables them to distinguish between prosocial and antisocial agents (e.g., Geraci et al., 2025; Hamlin, 2013a; Hamlin & Wynn, 2011; cf., Lucca et al., 2025; Schlingloff et al., 2020). For instance, in a classic study, Hamlin et al. (2007) showed that 6- and 10-month-old infants preferred a Helping character that helped a protagonist reach the top of a hill over a Hindering character that prevented the protagonist for reaching the top of the hill. As with arguments against an inborn number ability, it has been argued that these data may instead reflect the operation of simple associative processes (e.g., Scarf et al., 2012), a preference for imitative agents (e.g., Powell & Spelke, 2018), or some combination of both (e.g., Benton & Lapan, 2022). Finally, it has been claimed that infants as young as 5 months of age, if not earlier, understand that objects continue

to exist when hidden based in part on the finding that they show increased looking when a screen ostensibly rotates through a box (impossible event) than when it stops rotating once its reached the box (possible event; e.g., Baillargeon et al., 1985; Baillargeon, 1987; for conceptually similar findings, see Baillargeon, 1986; Baillargeon & DeVos, 1991). Others, however, have argued and showed through rigorous experimentation that infants' responses to the two test events can be explained by well-documented familiarity and novelty preferences (e.g., Bogartz et al., 2000; Hunter & Ames, 1988; Hunter et al., 1983). For example, infants who were not given enough time to process a familiarization event in which the screen rotates 180° in the absence of a box looked longer at the perceptually more familiar but physically impossible test event than at the perceptually more novel but physically possible event. In contrast, infants who were given sufficient time to process the familiarization sequence showed increased looking to the perceptually novel test event.

As mentioned at the outset of this section, lean accounts differ in their specifics but share a core commitment: conceptually rich, innate knowledge and theory-like inferences based on domain-specific mechanisms are not necessary to explain infants' looking behaviors in VoE studies. Instead, these accounts maintain that such behaviors may reflect factors such as how strongly a stimulus was encoding and whether that stimulus is familiar or novel after being encoded, and the operation of domain-general learning mechanisms applied to low-level, semantically neutral features (i.e., arbitrary perceptual inputs with no inherent meaning or conceptual significance).

A *new* lean proposal for the origins of infant action understanding

As the discussion to this point has shown, there have been many attempts to explain infants' looking behaviors in VoE studies from a lean perspective. However, efforts to explain infants' action understanding and processing from the same perspective remain relatively scarce. A notable exception to this is a computational investigation by Overwalle (2010). One of the goals of this study was to explain infants' looking behaviors in Gergely et al. (1995). Overwalle (2010) argued that infants learn about what makes an action efficient (or inefficient) based on prior experience either observing or engaging in those actions rather than on a priori notions of efficiency or domain-specific mechanisms. Thus, infants might learn to see jumping as the most efficient action to perform in the presence of a wall if, each time an agent jumps, a wall is present. In contrast, they might learn to see straight motion paths as most efficient if agents engage in this action whenever an obstacle is absent. Overwalle (2010) implemented this idea in a recurrent connectionist model and showed that it could explain infants' looking responses across the Experimental and Control conditions in Gergely et al. (1995). Overwalle (2010) interpreted this finding to mean that knowledge about what makes an action efficient can be learned from experience.

Although Overwalle's (2010) results suggest putatively that action efficiency can be learned, caution should be exercised before accepting this conclusion for at least three reasons. The first is that because the model did not possess one (or more) hidden layers, it could not develop internal representations. This inability to develop internal representations is problematic because it meant that the model could not make use of any prior, "real-world" experience. Yet, infants probably do enter studies like that by Gergely et al. (1995) with substantial experience processing others' actions, and this experience may interact with more general processes during the experiment to influence how infants process the various action events shown to them. Thus,

if Overwalle (2010) is correct that infants learn what makes an action efficient by observing actions in the real world, then it is crucial to demonstrate that such experience is carried over to influence infants' looking behaviors in Gergely et al. (1995).

A second limitation concerns how Overwalle (2010) simulated the Experimental and Control conditions in Gergely et al. (1995). It will be recalled that infants in both conditions saw a short wall during habituation; what distinguished the conditions was whether the first circle jumped over the wall (Experimental condition) or beside it (Control condition). In contrast, such a wall was not included in Overwalle's (2010) simulation of the habituation event in the Control condition; it only appeared in the Experimental condition. Thus, the model's success may simply reflect a failure accurately to simulate the Control condition. A final limitation is that the only action understanding study that Overwalle (2010) focused on was that by Gergely et al. (1995). It thus remains an open question whether the same model can account for findings beyond this one.

With these limitations in mind, the goals of this paper were twofold. The first was to present an alternative lean account of infant action understanding that can explain several key findings in the literature. Although the focus here is on infant action understanding, it is my contention that the proposed account and the findings from the simulations reported below speak to a much broader point: domain-general learning mechanisms can account for infants' behavior in VoE studies in general and their developing action knowledge in particular. My central argument is that infants' performance in action understanding VoE studies can be explained by the real-world, third-person experience of observing and processing others' action as well as by online associative learning. Specifically, infants' online behavior in these studies reflects and is shaped by extensive experience observing movement in the real world together with an associative learning mechanism that operates over surface-based and semantically neutral

kinematic features of movement. In the classic study by Gergely et al. (1995), for instance, infants may have formed an association between the arced motion path and the barrier based on repeated pairings of the two over the course of the habituation phase. Crucially, this link may come to be particularly strong when the trajectory occurs over rather than beside the barrier. This is because when the jumping action occurs *over* the barrier, the curved motion and the barrier occupy the same region of the screen and thus can be simultaneously fixated. This may have two consequences. First, it makes the barrier visually more salient. Second, this increased salience may enhance the likelihood that an associative link is established between the barrier and the curved motion. In contrast, when the curved motion occurs *beside* the barrier, as is commonly the case in these studies, more effort is required to form this link because infants cannot simultaneously fixate the curved motion and the barrier. Consequently, for infants to form an association between the barrier and the curved motion in this configuration, they must repeatedly shift their attention from the jumping action at the center of the screen to the barrier positioned off to the side. This constant need to shift their attention may lead to a weaker association between the curved motion and barrier, particularly because the barrier receives less of a salience boost in this configuration than when it shares the same region of the screen with the curved motion. This mechanistic account helps to explain why infants in Gergely et al.'s (1995) Experimental condition—but not the Control condition—looked longer at the barrierless curved motion event than at the barrierless straight motion event: Because it is the case that the curved trajectory becomes strongly associated with a previously seen barrier only in the Experimental condition, the barrier's absence at test violates infants' learned expectations and thereby elicits increased looking to the curved motion event relative to the straight motion event.

The present simulation studies

Given the admittedly speculative nature of this proposal, it was important to show that it could in fact capture existing data on infant action understanding. To achieve this, the proposal was instantiated in a connectionist model (see Singh et al., 2025 for a gentle introduction). Critically, the model was not developed to simulate any one VoE study on infants' action understanding. Instead, the goal was to capture a broad set of findings that, to this point, have been taken as evidence of conceptually rich knowledge about action and specialized processes that subserve that knowledge. By simulating a broad set of findings rather than one or two specific findings, it was possible to demonstrate the explanatory breadth of the current account in particular and the sufficiency of lean accounts in general. The following findings were the focus of the present simulations:

1. Infants look longer at inefficient test events than at efficient ones if the jumping action occurs over rather than besides a barrier (e.g., Gergely et al., 1995).
2. Infants' looking behaviors towards inefficient and efficient actions does not depend on whether the characters possess animate features (Csibra et al., 1999; Sodian et al., 2004).
3. Infants can infer external constraints from an agent's actions and goals as well as an agent's goal from its actions and external constraints (e.g., Csibra et al., 2003).
4. Infants represent efficiency continuously (e.g., Liu & Spelke, 2017).

It should be clear from the list above that all the findings derive from studies that used some version of the Gergely et al. (1995) VoE method (henceforth, the Gergely paradigm). The rationale for focusing on findings from these studies is that few paradigms have been used more extensively to study infant action understanding than this one. Another reason is that it was

possible to test the explanatory scope of the current account with the same basic architecture. Although the present simulations focused on findings from studies based on the Gergely paradigm, I see them—and the theoretical proposal they implement—as speaking to a much broader point: Domain-general learning mechanisms may suffice to explain infants' behaviors in VoE tasks, on the one hand. and the development of their action understanding, on the other.

In the remainder of the paper, I introduce the model architecture used in each simulation, along with details about the pretraining experience that all models received prior to their participation in a specific simulation. I then present four simulations, each corresponding to one of the aforementioned key findings. Given that the focus here was to show that a model that implements the current perspective can capture data from studies that use the Gergely paradigm, each of the studies that illustrate one of the previous findings will be modeled only at the most general level, and all will be construed with the context of that paradigm. This means that details deemed nonessential for assessing the implications and consequences of the present theory were omitted (e.g., the perceptual overlap between objects, the exact lighting conditions of a study, the full range of actions that infants might plausibly encounter in everyday life, among others).

General Method

Network Architecture

The current series of simulations used a three-layer, feedforward autoassociative and autoregressive artificial neural network model (e.g., Mareschal et al., 2000). Networks were trained with backpropagation in cross-entropy error (e.g., Rumelhart et al., 1986). The hidden units used a RELU activation function to prevent the computed gradients from vanishing across time steps; the output units used a sigmoid activation function. Gaussian noise was added to the

input to reflect the assumption that infants process incoming sensory signals noisily. The value of this parameter was set to 0.375 (M = 0, SD = 0.375). In terms of the architecture itself, the goal of the autoassociative part of the network was to compress the input via the hidden units and then reconstruct it at the output layer. The goal of the autoregressive part of the network (described in more detail below) was to predict a given object's position at the next time step.

The model consisted of three layers (Figure 1). The input and output layers each contained 11 groups of units. The groups in the input and output layers were identical, and all groups used a localist coding scheme (i.e., a single unit was used to represent a given concept or feature). Localist representations were adopted to ensure that no two objects were more similar to each other than any other object. It is worth noting that the values of units that are unspecified in the text are assumed to be 0 (corresponding to the unit being "off"). In contrast, the hidden layer consisted of a single group of units and the representations developed over these units were largely distributed (i.e., multiple units coded for a given concept or feature, and the same unit could participate in coding different concepts or features; Hinton, 1984). The hidden layer consisted of 50 units.

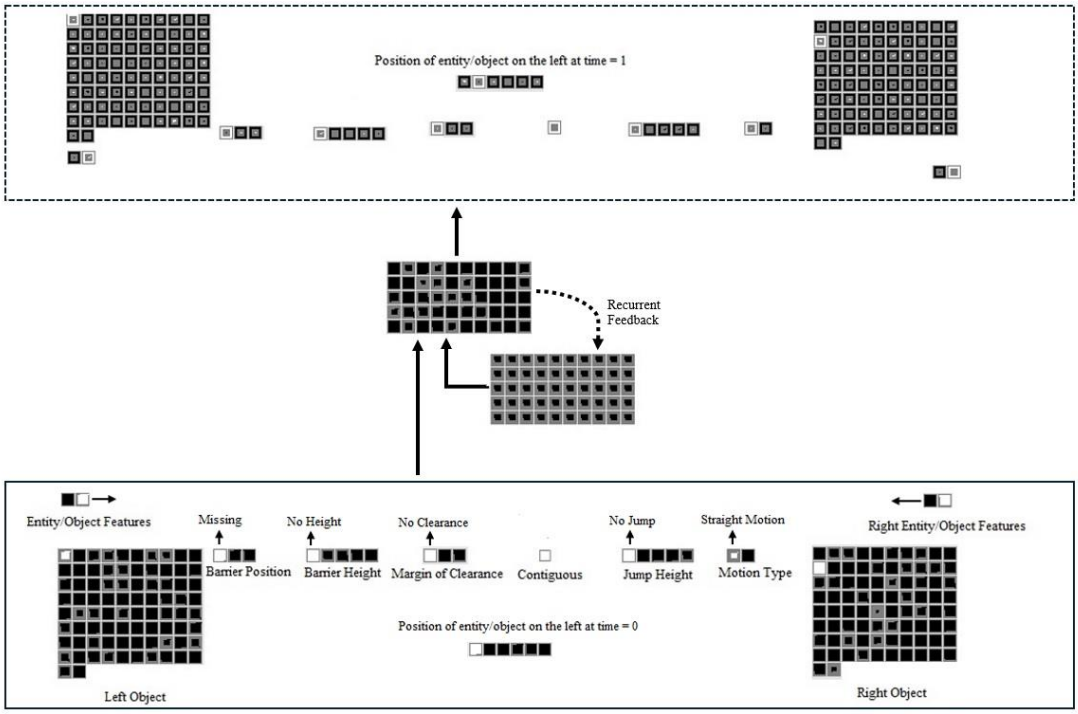

Figure 1. Schematic of the untrained model architecture used in the current simulations. Of note is that the input and output groups are identical.

The first two input groups were Object Left and Object Right groups. Copies of each input group also appeared on the output side of the model. Two different object groups were used because studies that used the Gergely paradigm tended to use two objects or entities. Crucially, each group consisted of 82 units. One of these groups was located on the network's "left side" (hence, Object Left); the other group was located on the network's "right side" (hence, Object Right). A single unit was used to represent a given object. Eighty objects were used for training; the remaining two were used for testing. Units that coded for objects present in a scene were turned "on"; that is, their values were set to 1.

The next two input and output groups were two-unit Animate Feature Left and Animate Feature Right groups. The Left-Right designation, as before, denoted on which side of the

network the group appeared. The first unit in either group was the Animate Feature unit; the second unit in either group was the Inanimate Feature unit. This meant that the object on the left or object on the right possessed an animate feature if the first unit of the corresponding right or left Animate Feature group had a value of 1. In contrast, the object possessed an inanimate feature if the second unit in either of these groups had a value of 1.

The next group of input and output units was a six-unit Motion group. Although the network could "see" at any point in time two agents or objects, a single motion group was used. On each timestep, a single bit in the motion layer was active according to the first object's or entity's position. In particular, at time $t = 1$, the activation value of the 1st (or leftmost) unit in the motion group in the input layer was set to 1 (the activation values of the remaining units were set to 0), and the network's task was to activate the second unit (from the left) in the corresponding motion group in the output layer. At time $t = 2$, the activation value of the second input unit in the motion group was set to 1 (all other values were set to 0), and the network's task was to turn on the 3rd output unit (from the left) in the corresponding output group. Movement continued in this manner until the 5th input unit was activated in the input layer. The network's job at this point was to activate the 6th unit in the output layer. It should be noted that although the described motion resembled that of a straight line, it is better to think of this group as capturing whether the left entity or object moved as well as its location in network space relative to the second object or entity (e.g., closer or further away from it); that is, this group represented simply *that* an object was moving but not the nature of that movement. The type or nature of the left object's movement was instead captured by a separate group of input and output units called the Motion Type group. The first unit in this group was the Straight Motion unit; the second unit was the Curved Motion unit. This group specified the type of movement represented in the

Motion group. Movement and its type were separated in this way to simplify later analyses of the model's internal dynamics. Nonetheless, it is acknowledged that these two aspects of motion are unlikely to be perceived independently in the real world, and a more realistic model would need to simulate curved and straight motion as single, dynamic activity patterns. In general, it was critical to simulate actual movement because the findings on which the present simulations were based derived from studies that habituated infants to events in which objects move. A second reason it was important to simulate motion was that infants' emerging action understanding stems, at least in part, from processing dynamic actions as they unfold in the real world.

The remaining five input and output groups were a five-unit Barrier Height group, a five-unit Jump Height group, a three-unit Barrier Location group, a third-unit Clearance group, and a one-unit Contiguous group. The five units in the Barrier Height group represented progressively larger barriers (i.e., Barrier Height 0 through Barrier Height 4). Similarly, the five units in the Jump Height group represented increasingly higher jumps (i.e., Jump Height 0 through Jump Height 4). Note that Barrier Height 0 corresponds to a barrier with no height. The same is true for Jump Height 0; that is, Jump Height 0 corresponds to the lack of a jump. In the simulations reported below, jump heights mostly aligned with barrier heights (though see Simulation 3). The first unit in the Barrier Location group corresponded to the Barrier Missing unit, whereas the remaining two units corresponded, respectively, to Barrier Middle and Barrier Side units. As the name of the group implies, this group encoded the location of the barrier. In terms of the three-unit Clearance group, the first unit corresponded to no clearance (which occurred when the agent did not jump), the second to minimal clearance (which occurred when an agent jumped just high enough to clear the barrier), and the third to large clearance (which occurred when an agent jumped much higher than needed to clear the barrier). This group encoded the vertical distance

between a jumping agent and the barrier it jumped over. Finally, the single unit in the Contiguous group encoded whether two objects came to rest next to one another following a reaching or jumping event. This unit was turned on (i.e., its value set to 1) only if the first object came to rest next to the second object.

Although I use semantically rich terms to describe the features in the model, this is not meant to suggest that infants represent these features in a conceptually or semantically meaningful way. These terms are used solely to help readers understand the model's structure. As is assumed to be true of infants, the present models similarly process these features as low-level perceptual properties rather than in terms of their conceptual meaning. Nonetheless, a potential objection is that because the features that the model processes are stipulated in the input rather than discovered as an infant might in the real world, the learning task differs in important ways from the one that infants face in that world. Specifically, by deciding which features to implement, the aspects of the input that the model attends to have been prespecified. There are two responses to this potential criticism. First, it has been shown that when some features are made more salient than others—as is the case here—models like that used here can discover autonomously which features to focus on, and this is true even if the salient and less salient features are perfectly correlated (e.g., Benton, 2024; Rakison & Lupyan, 2008; Singh et al., 2025). Second, it is assumed here that attention to these low-level features as well as sensitivity to the associative relation among them is sufficient to explain infants' looking behaviors in Gergely-style VoE studies. If a model exposed to these features can capture infants' responses in studies on infant action understanding, this would suggest that infants' performance in these studies may reflect sensitivity to similar features. At the very least, such a demonstration would suggest that sensitivity to these surface-level features and the relations among them is sufficient

to explain infants' looking behaviors in infant action understanding studies and that abstract knowledge that is underpinned by specialized processes might not be necessary.

Pretraining

Given that the current theoretical proposal assumes that infants' real-world experiences with action influences their looking behavior in action understanding studies, it was important to approximate this experience in the model. As stipulated by the account, networks experienced two different kinds of events during pretraining. In Straight Path pretraining events, one of the objects "followed" a linear path to reach the other one; that is, one of the 82 units in the object group on the left and one of the 82 units in the object group on the right were turned on, along with the Straight Motion unit in the Motion Type group. Other units that were turned on for all straight motion events were the Barrier Height 0 unit in the Barrier Height Group, the Jump Height 0 unit in the Jump Height group, the Barrier Missing unit in the Barrier Location group, the No Clearance unit in the Clearance group, and the Contiguous unit in the Contiguous group. In the Curved Path pretraining events, one of the objects "jumped over" a short barrier to reach the other object. Units that were turned on for this event were the Barrier Height 1 unit in the Barrier Height group, the Jump Height 1 unit in the Jump Height Group, the Barrier Middle unit in the Barrier Location group, the Contiguous unit in the Contiguous group, and the Curved Motion unit in the Motion Type group.

To instantiate the assumption that things with or without animate features can follow curved or linear motion paths, half of the objects within each type of event possessed animate features (i.e., the Animate Feature unit in the Animate Feature groups was activated) and half possessed inanimate features (i.e., the Inanimate Feature unit in the Animate Feature groups was

activated). In addition, the pretraining events instantiated the assumptions that curved motion is more salient than straight motion, barriers are more visually salient when motion occurs over them than beside them, and agents jump just high enough to clear barriers. The first assumption was implemented by assigning a higher learning rate to the incoming and outgoing weights of the Jump Height units during curved motion events compared to linear motion events. Specifically, for curved motion pretraining events, the learning rate was set to 0.375. The learning rate for curved motion was set even higher to 0.475 during habituation based on the assumption that, unlike in the real world with its many distractions, infants in this phase are likely focused exclusively on the presented motion events. To instantiate the second assumption, the learning rate on the incoming and outgoing links to the Barrier Height group was set to 0.375 during the pretraining phase, to 0.475 whenever curved motion occurred above it during habituation, and to 0.0001 whenever curved motion occurred besides it. To further decrease the salience of the barrier when it occurred beside the motion, weight decay was added to the incoming and outgoing weights to the Barrier Height group. The value for this parameter was 0.01. Learning rate is a parameter that controls the speed at which the weights in the network adapt; larger learning rates lead to more rapid weight changes and hence faster learning. In contrast, weight decay controls how “big” the weights are allowed to grow; lower values for this parameter correspond to larger weights. Typically, learning rate and weight decay are values less than 1. The decision to implement salience through the learning rate and weight decay aligns with past connectionist modeling work (e.g., Benton, 2024; Singh et al., 2025). The last assumption was instantiated by turning on the Small Clearance unit in the Clearance group when agents jumped over barriers.

Across all four simulations, the pretraining phase lasted 800 epochs. The learning rate for the non-Jump Height and non-Barrier Height connections was 0.1. Crucially, the nature of this pretraining experience was identical for all networks, regardless of simulation. The habituation phase always followed the pretraining phase and lasted between 2 and 12 epochs in step sizes of 2. Different numbers of epochs were used to ensure that the results reported below held for different values of this parameter. Accordingly, data from all five habituation durations are shown below for each of the simulated findings.

The dependent measure used in these simulations was network (cross-entropy) error (Hinton, 1989). Greater error corresponds to longer looking. Using error as a proxy for infant looking is typical in simulation studies like these (e.g., Benton & Lapan, 2021; Rakison & Lupyan, 2008). To understand why this is the case—that is, why it is appropriate to conceptualize network error as being positively related to infant looking time—it is important to consider the relation between infant habituation and learning in standard autoassociative models like that used here. According to one influential model (Sokolov, 1963), habituation is a process in which learners gradually adjust their internal representation of an external stimulus until it matches the external stimulus in some way. While there is a discrepancy between the internal representation of the external stimulus and the external stimulus itself, learners will continue visually to fixate the external stimulus. Only when this discrepancy is resolved do learners disengage from the external stimulus and turn their attention elsewhere. In a similar vein, autoassociative models learn to reproduce at the output layer the input (akin to the external stimulus) that is presented to the model. These networks do this by gradually adjusting their hidden representations (akin to the model's internal representation) until those representations are sufficient to reproduce that input at the output. As was true for infant habituation, network

error will remain high (akin to the network continuing to fixate the external stimulus) until the discrepancy between the model's representation of the external stimulus at the hidden layer matches the external stimulus in some way. Only at this point is the network ready to process (i.e., fixate) a new external stimulus.

All simulations were run using the Lens neural network simulator (Rohde, 1999). All training and testing files, including the data from the simulations, are available for download at https://osf.io/9eqpf/?view_only=62f752417a9141c7b0509a998039ad13.

Simulation 1

Previous research found that infants looked longer at an inefficient event than at an efficient one after habituating to an event where an animate-like object either jumped or reached over a barrier to reach another animate-like object (e.g., Gergely et al., 1995; see also Sodian et al., 2004). However, infants attended equally to both events if, during habituation, the barrier was located alongside rather than beneath the jump or reach. Simulation 1 tested whether the current model (and thereby, the current theoretical proposal) could account for this finding.

Networks were assigned either to the Experimental condition (N = 20) or to the Control condition (N = 20). A total of 240 networks were tested—20 networks for each training epoch and condition. The habituation event in the Experimental condition mirrored the Curved Path pretraining events with two key exceptions. First, novel objects with animate features were used. Second, the fourth unit rather than the first unit was turned on in the Barrier Height and Jump Height groups. These differences ensured that networks' processing during the habituation phase was not influenced by prior experience with either the pretraining barrier height or specific objects or entities. Networks were then shown a straight path event (Efficient test event) and

curved path event (Inefficient test event). Both events were shown without the barrier and used the same objects from habituation.

Results

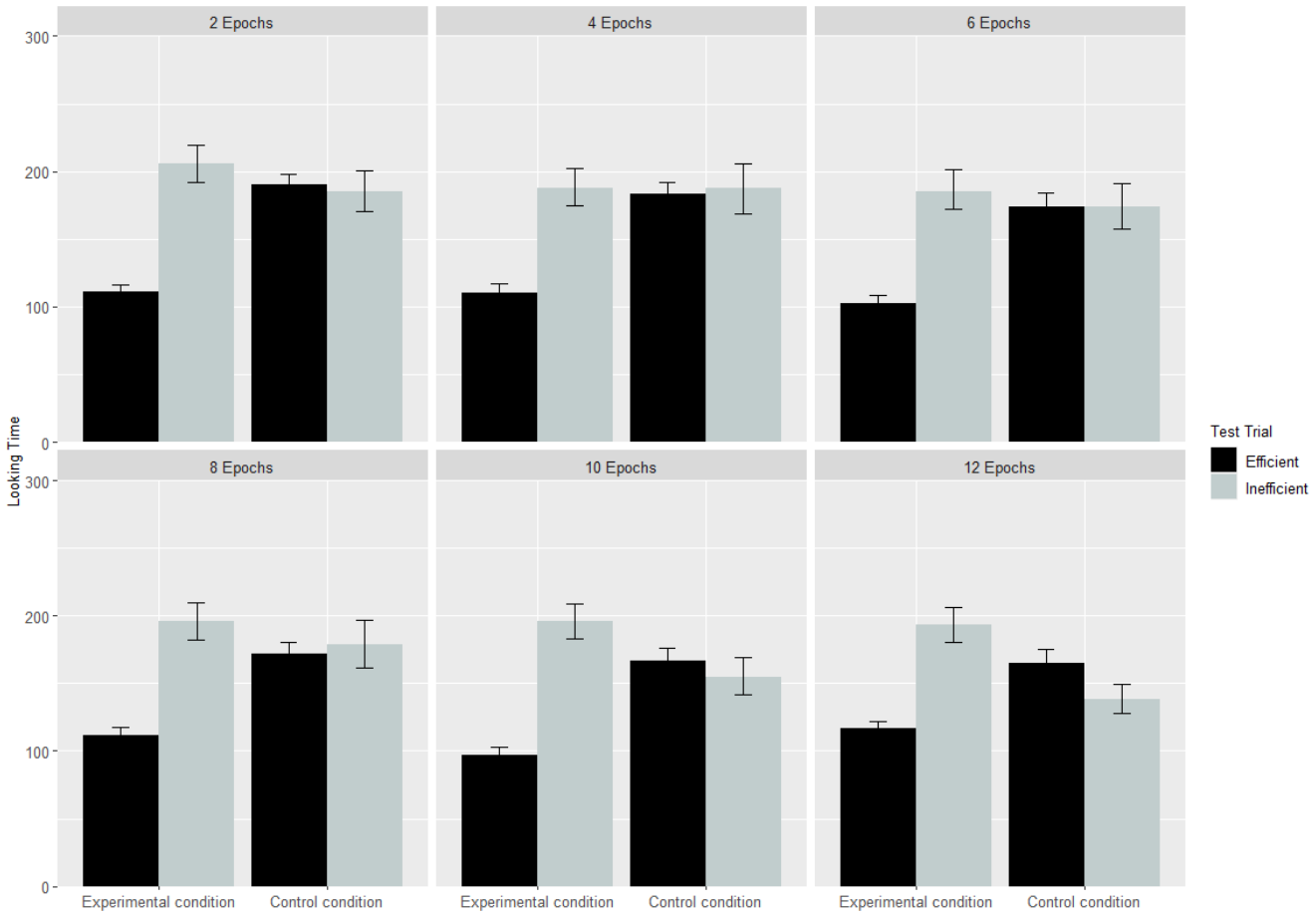

Figure 2. Networks' mean "looking time" (i.e., cross-entropy error) to the Efficient and Inefficient test events across conditions and training epochs.

As can be seen in Figure 2 above[1], networks in the Experimental condition consistently looked longer at (i.e., produce greater error in response to) the Inefficient test event than the Efficient test event. A different pattern emerged for networks in the Control condition. Specifically, it was the case that networks' pattern of looking across the different training epochs and test events was variable; that is, networks looked longer at the Efficient test event than at the Inefficient test event for some training epochs and looked longer at the Inefficient test event than at the Efficient test event for others. Overall, then, networks in the Control condition looked about equally at both test events.

Discussion

Previous research showed that infants look longer when agents pursue goals inefficiently than when they pursue goals efficiently but only if an agent's earlier actions were justified given external constraints. This finding and others have been interpreted to mean that infants' action understanding is underpinned by domain-specific mechanisms and principles that support conceptually rich knowledge and theory-like inferences about action, utility, cost, and efficiency. This simulation tested whether this conclusion was warranted; that is, it assessed whether a system that sees an equal amount of straight and curved motion during a pretraining or "real

[1] Inferential statistics were not used in any of the simulations for two reasons. First, because infant looking times and model error are measured on fundamentally different scales and exhibit different distributions, direct statistical comparisons between them were not appropriate. Second, the goal of the current simulations was not to show that the magnitude of the model's effects matched those observed in empirical studies. Instead, the aim was to demonstrate that the qualitative pattern of results—such as greater error to unexpected or inefficient events in some conditions but not others—aligned with the patterns observed in infant behavior. This emphasis on qualitative pattern matching over statistical inference is consistent with prior modeling work in developmental science (e.g., Benton, 2024; Mareschal et al., 2000; Mareschal & Schultz, 1999; McClelland & Thompson, 2007; Plaut & Vande Velde, 2017).

world" phase and that learned more strongly to associate curved motion with a barrier when the motion occurred over the barrier than when it occurred beside it could account for this finding. The results of this simulation provided a positive answer to this question: Networks, like infants, only looked longer at (i.e., produced greater error to) the Inefficient test event than at the Efficient test event if the barrier was beneath rather than beside the curved motion during habituation. This result supports the contention that infants' looking behaviors in VoE studies like that by Gergely et al. (1995) may result from seeing curved and linear action in the real and an associative learning mechanism that operates online during an experimental session to form links among perceptual features that vary in strength based on their perceptual saliency. However, this result alone cannot show the explanatory breadth of the current theoretical account. Simulation 2 was the first step in this direction. This simulation tested whether the current results depended on the presence of animacy features.

Simulation 2

Simulation 2 was loosely based on that by Csibra et al. (1999) and tested whether networks' behavior in Simulation 1 depended on the presence of animate features. This simulation was identical to Simulation 1 in all respects except for the fact that the objects used during the habituation and test phases did not possess animate features; that is, the Inanimate Feature unit was turned on in both Animate Feature groups.

Results

Figure 3 depicts networks' mean looking times to the Efficient and Inefficient test events for the Experimental and Control conditions across the training epochs. As can be seen in that

figure, networks assigned to the Experimental condition looked longer at the Inefficient test event than at the Efficient test event. This was true across all training epochs. In contrast, and as was true for Simulation 1, networks in Control condition showed an inconsistent pattern of looking to the test events across the training epochs—for some of the training epochs, networks looked longer at the Inefficient test event, whereas for other training epochs, networks showed the reverse pattern of looking. Overall, networks in the Control condition looked about equally at the Efficient and Inefficient test events. These results reproduced those observed in Simulation 1.

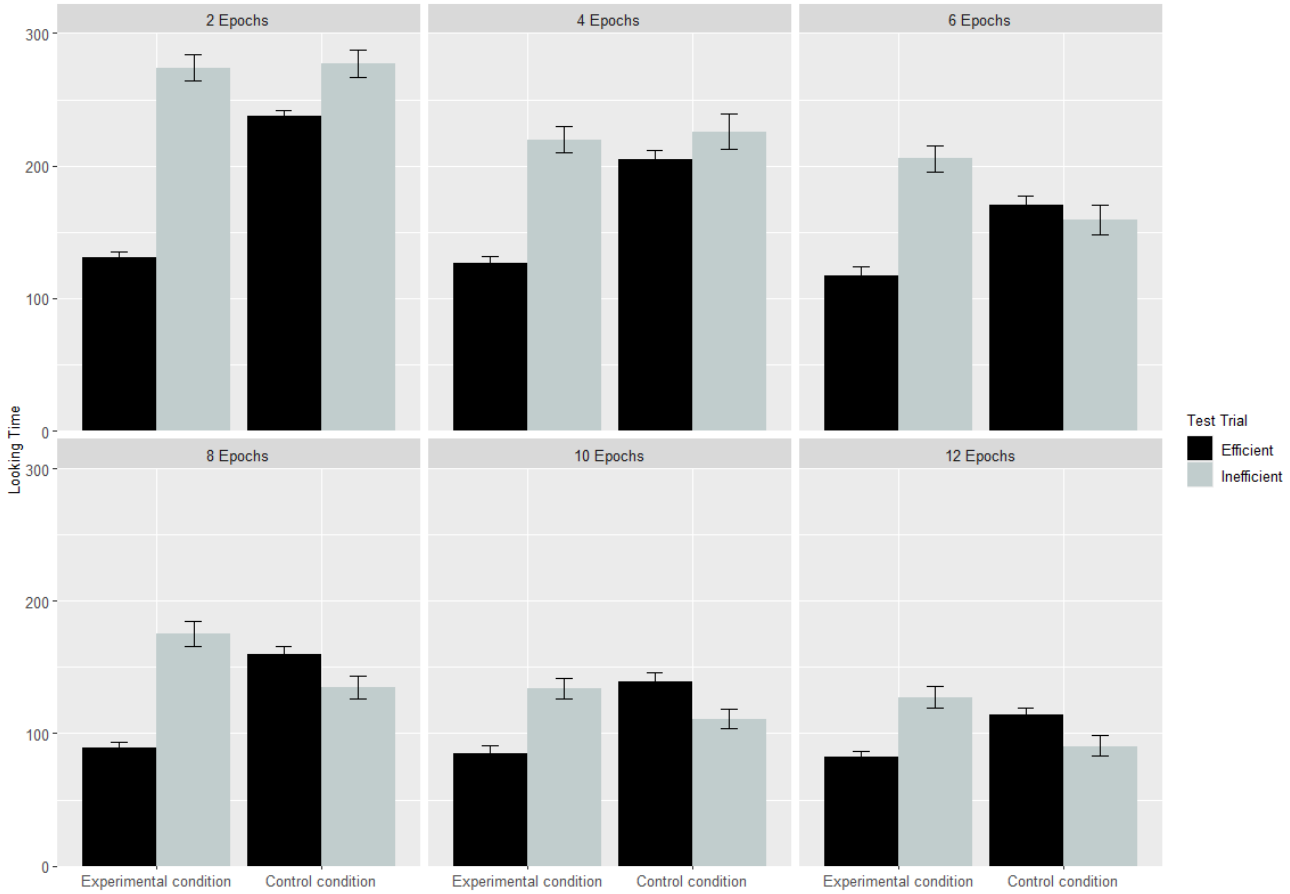

Figure 3. Networks' mean "looking time" to the Efficient and Inefficient test events in Simulation 2 across conditions and training epochs.

Discussion

Studies such as that by Csibra et al. (2003) showed that infants continue to distinguish between inefficient and efficient actions even if the entities involved in the action events do not possess animate features. This simulation examined whether an associative learning mechanism, combined with extended real-world experience observing curved and linear motion, can account for this finding. Specifically, the main aim of Simulation 2 was to determine whether the simulation results from Simulation 1 were due to the objects possessing animate features. The results of Simulation 2 showed that it was not necessary that the objects involved in the action sequences possess animate features. As was true for infants in Gergely et al. (1995) and Csibra et al. (1999), networks showed heightened looking at (i.e., increased cross-entropy error in response to) the Inefficient test event relative to the Efficient test event if earlier the agent jumped over a barrier than if the agent jumped beside it. This result provides additional support for the notion that infants' looking responses in Gergely-style studies can be explained by an associative learning mechanism, combined with experience with curved and straight motion in the real world.

The next simulation was designed to address a finding that represents perhaps the strongest evidence for a domain-specific mechanism such as a teleological stance: Infants can make inferences about one of the components of the model given information about other components. For example, infants can infer that an external constraint exists based on an agent's actions and its goals. Similarly, infants can infer an agent's goals from its actions and external constraints. Data that bear directly on this question come from a study by Csibra et al. (2003). In one of the experiments, 12-month-olds were habituated to an event in which a small circle entered the screen from the bottom left, moved straight through a gap between two bars, and then

disappeared from the screen. A larger circle, which entered the screen from the bottom right shortly after the small circle emerged, moved in the direction of the small circle as if chasing it. Because the gap between the two barriers was too narrow for the larger circle to fit through, it moved around the right bar before continuing in the small circle's last seen direction. Crucially, because the larger circle never reached the small circle, its goal—namely, to catch the small circle—had to be inferred. Infants were then shown two test events. In both test events, the gap between the two barriers was now wide enough for both circles to pass through. In the Incongruent event, the larger circle again moved around the right barrier; it passed through the gap in the Congruent event. Csibra et al. (2003) found that infants looked longer at the former event than at the latter. This was interpreted to mean that infants understood that the larger circle's goal was to catch the smaller circle.

In another experiment, infants were habituated to an event like that shown in the Experimental condition in Gergely et al. (1995) except that the barrier was hidden behind an occluder. However, infants could still see the second object and portions of the first object's curved motion path behind the occluder as it moved to reach the second object. Infants were then shown two test events without the occluder. The first event depicted the first object jumping over a now visible barrier to reach the second object (Obstacle test event). The second event was identical to the first except that the obstacle was missing (No-Obstacle test event). The results indicated that infants looked longer at the No-Obstacle test event than at the Obstacle test event. The authors interpreted this as evidence that infants can infer invisible constraints from actions and goals. Together, these findings were taken to mean that infants possess a teleological stance.

Simulation 3 tested whether the present theoretical perspective was sufficient to explain these findings. Such a demonstration would indicate that domain-specific mechanisms like a

teleological stance is not needed to account for infants' behaviors in studies like that by Csibra et al. (2003). It is worth noting that although the experiment that tested whether infants could infer goals from actions and constraints did not use the Gergely paradigm, it is straightforward to recast it within that paradigm. For example, the habituation event—which would mirror that shown in the Experimental condition of Gergely et al. (1995)—could be stopped right before the first object reaches the second object but after the first object clears the barrier. Because observers never see the first object reach the second object in this configuration, observers must infer that the second object is the first object's goal. Simulation 3 adopted this approach. The rationale for doing this is that the model architecture did not need to be changed.

Simulation 3

Simulation 3 tested whether the model used in Simulations 1 and 2—and thereby the theoretical assumptions it implemented—could also explain the finding that infants ostensibly can infer goals from constraints and behavior as well as constraints from goals and behaviors. Networks were assigned either to the Inferring Goals condition (N = 20) or to the Inferring Constraints condition (N = 20). The model used to simulate the Inferring Goals condition was identical to that used in Simulation 1 except that the Contiguous group was absent during habituation. Thus, the network could not know whether the first object came to rest next to the second object (the Inferring Goals scenario). The model used to simulate the Inferring Constraints condition was also identical to that used in Simulation 1 except that barrier information was missing; that is, the Barrier Height and Barrier Position groups were absent.

Results

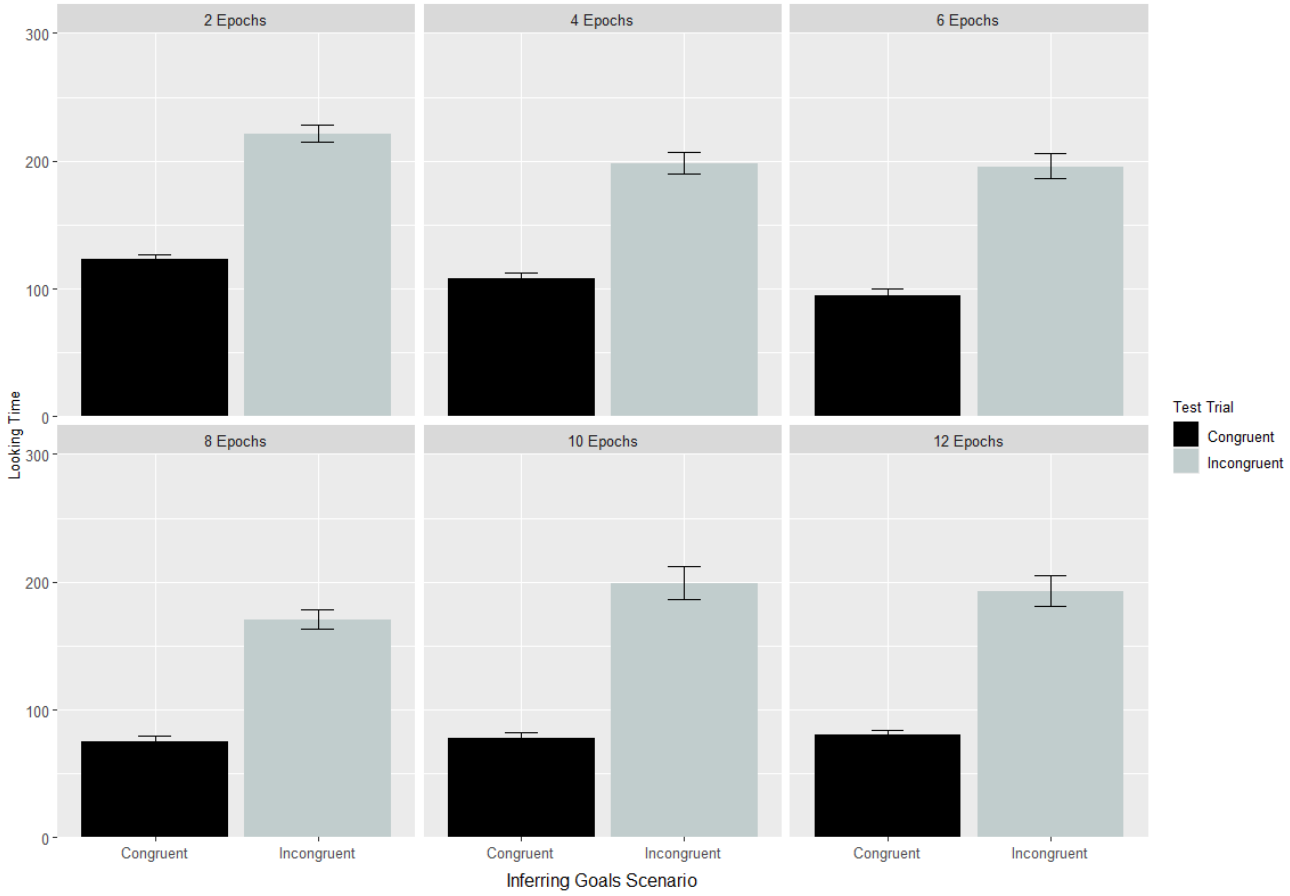

Figure 4. Mean looking times to the Congruent and Incongruent test events across the training epochs for networks assigned to the Inferring Goals condition.

Figure 4 shows the looking behavior of networks in the Inferring Goals scenario. It should be clear from the figure that networks in this condition looked longer at the Incongruent test event, or the test event in which the first object did not stop next to the second one, than at the Congruent test event, or the test event in which the first object came to rest next to the second one. This pattern held regardless of the length of the habituation period.

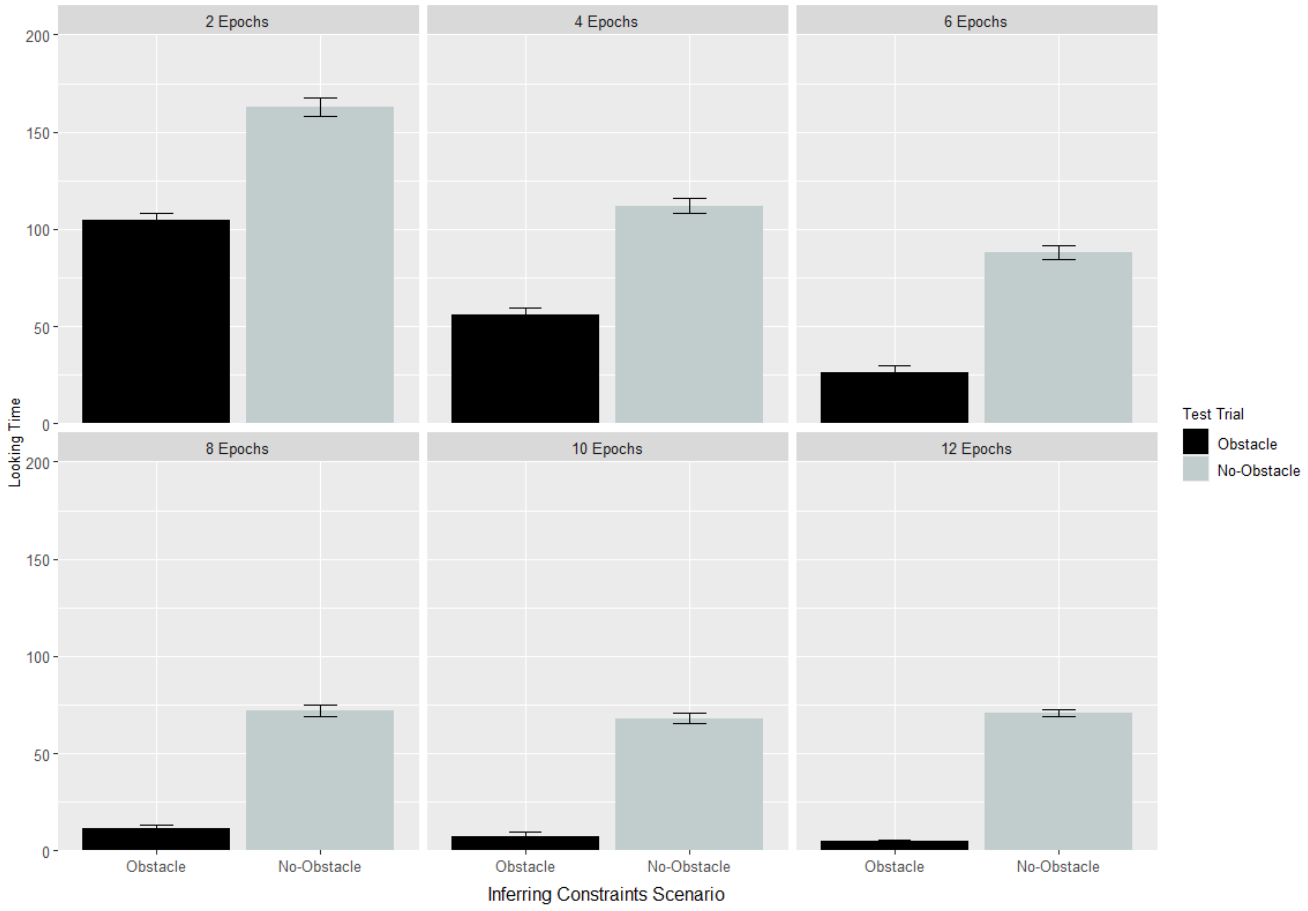

Figure 5. Mean looking times to the Obstacle and No-Obstacle test events across the training epochs for networks assigned to the Inferring Constraints condition.

Figure 5 shows the looking times of networks assigned to the Inferring Constraints condition to the two test events. As was true for networks in the Inferring Goals conditions, networks in this condition looked longer at the No-Obstacle test event, or the test event that lacked a physical barrier. than at the Obstacle test event, or the test event that contained a physical barrier, and this was true regardless of the length of the habituation phase.

Discussion

Perhaps one of the strongest pieces of evidence that infants possess specialized mechanisms for processing action, such as a teleological stance, is the finding that infants can infer different components of this model given others. For instance, the study by Csibra et al. (2003) suggested that 12-month-old infants can infer hidden physical constraints given information about an actor's goal and its actions as well as an actor's goal given information about the physical constraints in an environment and an actor's behavior in relation to those barriers. Although it may be tempting to conclude that infants possess such a stance based on findings like that by Csibra et al. (2003), the current simulation demonstrated that such an account is not necessary to explain this aspect of infants' action understanding. As with Simulations 1 and 2, Simulation 3 showed that prior experience with curved and linear motion as well as online associative learning that operates over low-level, semantically neutral surface features is sufficient to account for infants' looking behaviors.

The goal of the final simulation is to push the current theoretical proposal still further by evaluating whether it can explain another aspect of infant understanding: By 6 months of age, infants understand that actions can differ in their efficiency such that one event is more efficient (or inefficient) than the other (e.g., Liu & Spelke, 2017). This finding has been interpreted to mean that infants' action understanding is governed by something like a naïve utility calculus. Simulation 4 tested whether the current model could capture this finding.

Simulation 4

This simulation was inspired by Liu and Spelke (2017) and tested whether networks could distinguish between two events that differed in their relative costliness. Liu and Spelke (2017) habituated 6-month-old infants to an event in which a circle character jumped over a

barrier to reach a triangle character on the other side. A separate group of infants in a control condition were habituated to the same event expect that the jumping action occurred besides rather than over the barrier. Both groups of infants were then shown two test events. Crucially, unlike the test events used in other studies that employed the Gergely paradigm, which did not include a barrier, this study did include one. However, the barrier was much shorter than that shown during habituation. In the *low, efficient jump* test event, the agent jumped just high enough to clear this barrier. In the *high, inefficient jump* event, the agent jumped as high as it did to clear the taller barrier during habituation. Liu and Spelke (2017) predicted that infants should look longer at the second test event than at the first one if (a) they represent efficiency (or cost) continuously, (b) expect agents to engage in the least inefficient and costly action when pursuing goals, and (c) an agents' earlier costly action is justified (such as in the experimental condition but not in the control condition). This is what they found.

Networks in this simulation were habituated to an event in which one object with animate features jumped over a large barrier to reach another object with animate features. The large barrier was instantiated by turning on the Barrier Height 5 unit in the Barrier Height group and the Jump Height 5 unit in the Jump Height group. Networks were then tested with a High Jump test event and a Low Jump test event. The High Jump test event was identical to the habituation event (e.g., the jump height remained the same) except that it used a low barrier. This barrier was implemented by turning on the Barrier Height 2 unit in the Barrier Height group. Thus, the first object jumped much higher than needed to clear the now much shorter barrier. The Low Jump test event was identical to the High Jump test event except that the height of the first object's jump corresponded to the height of the now much shorter barrier (i.e., the Jump Height 2 unit was turned on).

Results

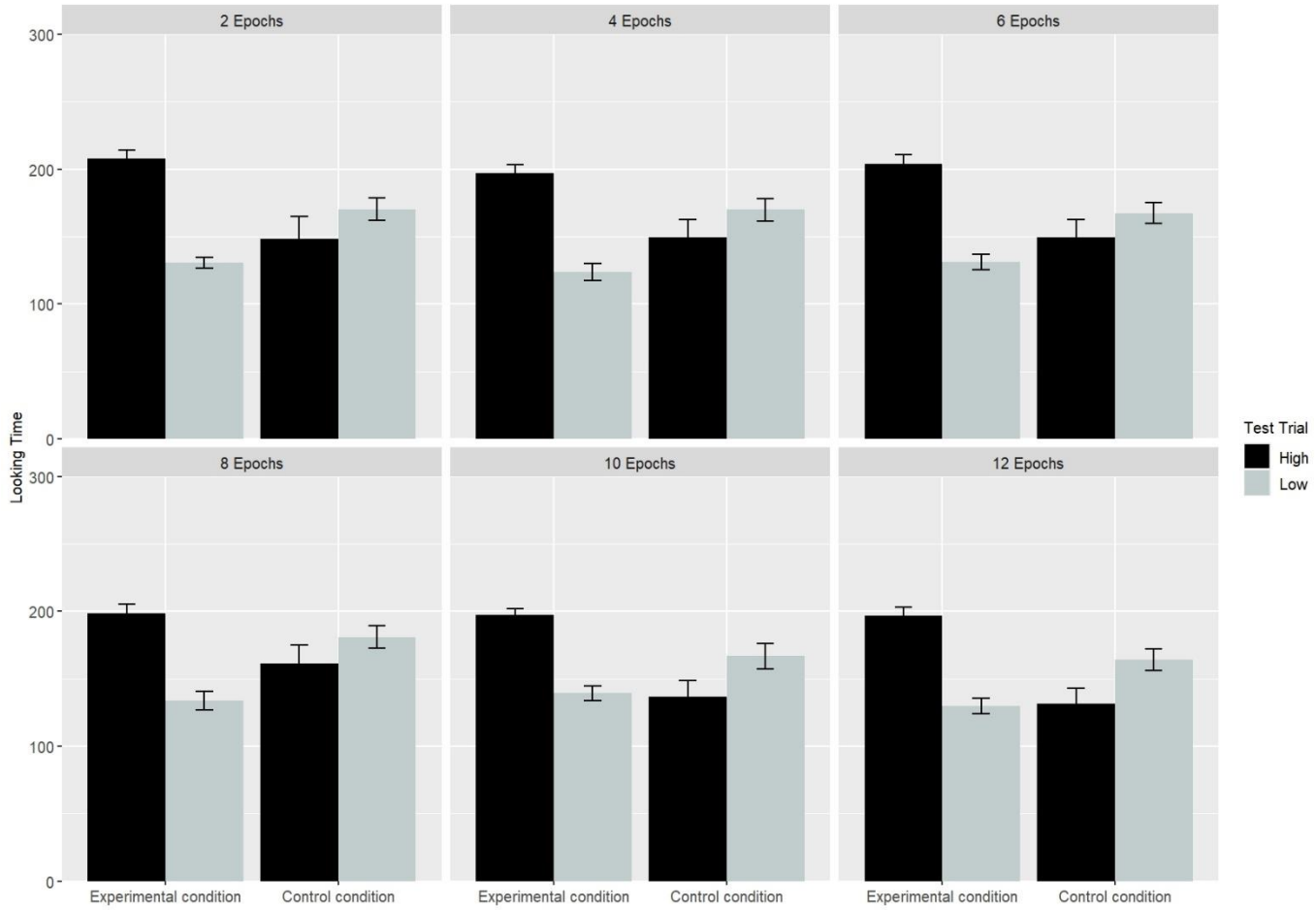

Figure 6. Mean looking times to the Efficient and Inefficient test events across the training epochs and conditions.

Figure 6 depicts networks' mean looking times to the Efficient and Inefficient test events across training epochs and conditions. As shown in the figure, networks in the Experimental condition looked longer at the high, inefficient test event than at the low, efficient test event. In contrast, and as was also true for the infants in the Control condition in Liu and Spelke (2017), networks showed a much slighter but still consistent tendency to look longer at the low, efficient test event than at the high, inefficient test event.

Discussion

The main aim of Simulation 4 was to capture the finding that infants look longer when agents engage in superfluously costly action than when those same agents engage in minimally costly action (e.g., Liu & Spelke, 2017). The results of this simulation showed that networks also looked longer at a more costly event than they did at a less costly one. This finding is significant because neither the current associative learning account nor the model that instantiated it implements a notion of cost, utility, efficiency, or action or domain-specific mechanisms such as a NUC or a teleological stance. This suggests that such principles and domain-specific mechanisms are not necessary to account for infants' looking behaviors. Instead, these results demonstrate that prior experience with different kinds of actions and an associative learning mechanism that is sensitive to and detects the relation between and among the low-level, surface features in action events may be sufficient to explain infants' looking behaviors in studies like that by Liu and Spelke (2017), among the others discussed in this paper.

Although the current series of simulations make a strong case that an associative learning mechanism is sufficient to account for infants' looking behaviors in studies that assess their action understanding, they do not—and indeed cannot—demonstrate that infants lack specialized mechanisms for processing actions. Similarly, the present simulations cannot speak to whether associative learning underpins infants' processing of action sequences in the real world. Instead, the present simulations serve as a proof of concept that associative learning alone is sufficient to explain infants' looking behaviors in VoE studies that assess their action understanding.

The simulations to this point show that associative learning can explain a number of findings in the literature on infant action understanding. But *that* associative learning, combined

with real-world experience with curved and linear actions, reproduces infants' looking behaviors says little about the precise nature of that learning. Put plainly, it is unresolved how associative learning operated in the model to enable it to capture infants' looking behaviors in some Gergely-style studies.

Unpacking the mechanism by which the model produced a match to the behavioral data

The simulations thus far suggest that real-world experience with straight and curved motion, combined with online associative learning, can explain infants' looking behavior in VoE studies that use the Gergely paradigm. However, the fact that associative learning mechanism captured the current set of findings is silent about how this process enabled the model to do so. Yet, elucidating how the model captured infants' looking behaviors is important because it can provide meaningful insight into how associative learning might operate in infants' minds to support action processing. One way to get insight into how the current model was able to reproduce infants' looking responses through associative learning is to consider how it processed and linked the various surface features across the different phases of the simulation. Given that the study by Gergely et al. (1995) laid the foundation for this line of research, I provide a detailed account of how the model reproduced infants' looking behavior in that study. I conclude this section by briefly discussing how those principles extend to capture the remaining findings.

Processing in the model begins during the pretraining phase. At the beginning of this phase, the network is presented with an action event. This event is encoded as a pattern of activation across different input groups (e.g., Object Left and Right, Barrier Location, Barrier Position, etc.). These events correspond to situations in which one object or animate entity moves along either a straight or curved path to reach a second object or animate entity. This

pattern of activation is then propagated to the hidden layer via the weights that connect each of the input groups to the hidden layer. At the beginning of training, this pattern of activity does not capture or reflect anything meaningful about the input to the model. This is because the weights themselves are small, random values. Despite the fact that this hidden activity has not encoded anything meaningful in the input, it is then propagated to, and elicits a pattern of activity over, the output groups in the output layer. This observed pattern of activation is then compared to the desired or target pattern of activation. Any discrepancy between the two patterns is treated as error, and the network uses this error to adjust the weights throughout the network.

Crucially, the pattern of error that the linear motion events produce over the output groups is different than that produced by the curved motion events. One consequence of this is that the weight changes will be different for the two action events; that is, the distinct error signals will cause the network weights to move in different directions for both events. Over time, these distinct weight changes lead to distinct and now meaningful hidden representations for both types of motion. Specifically, the network will develop a representation at the hidden layer that not only encodes information about motion but that reflects learned indirect associations between and among specific surface features. These associations are indirect because they are "routed" through the hidden layer. As such, features in the input (e.g., Curved Motion) will indirectly activate at the output layer the features with which they were associated during training (e.g., Jump Height 1). The fact that these associations had 800 epochs to develop during pretraining—combined with the fact that many of the same features reappear during the experimental phase—plays an important role in explaining how the network responds to the test events in these simulations.

For example, consider the habituation and test events shown to networks in the Experimental condition of Simulation 1. Recall that during this phase, networks are shown an event in which a novel object with animate features jumps over a barrier to reach another object on the other side. Importantly, two of these features are novel; the remaining two are familiar. The novel features relate to the height of the barrier (Barrier Height 4) and the height of the jump (Jump Height 4). The familiar features relate to the position of the barrier (Barrier Middle), the nature of the jumping action (Curved Motion), the size of the gap between the jumping agent and the barrier (Clearance Small), and the fact that the agent stops next to its goal (Contiguous). At the beginning of the habituation phase, the familiar features activate indirectly the features that they occurred with during the pretraining phase. By contrast, because the novel features were not experienced during the pretraining phase, new associations must be formed during habituation. Here, the network must learn that the new jumping action and the taller barrier are connected. Although the habituation phase is shorter than the pretraining phase, the network has little difficulty learning these associations while simultaneously preserving old ones. Consequently, if one (or more) of these features is presented to the model as input in either of the test events, it will activate—again via the hidden layer—features that it was associated with during the pretraining and habituation phases. The reason the network can learn these new associations and incorporate them into the old ones within the short timeframe of the habituation phase is because the salience of the Barrier Height 4 and Jump Height 4 units were heighted relative to that during the pretraining phrase. This reflected the assumption that barriers are visually more salient when they occur beneath jumps, as well as the assumption that jumps are intrinsically salient or attention grabbing. It will be recalled that this assumption was implemented by increasing the

learning rate on the weights to and from these units. Thus, salient features can be learned quickly in a short period of time.

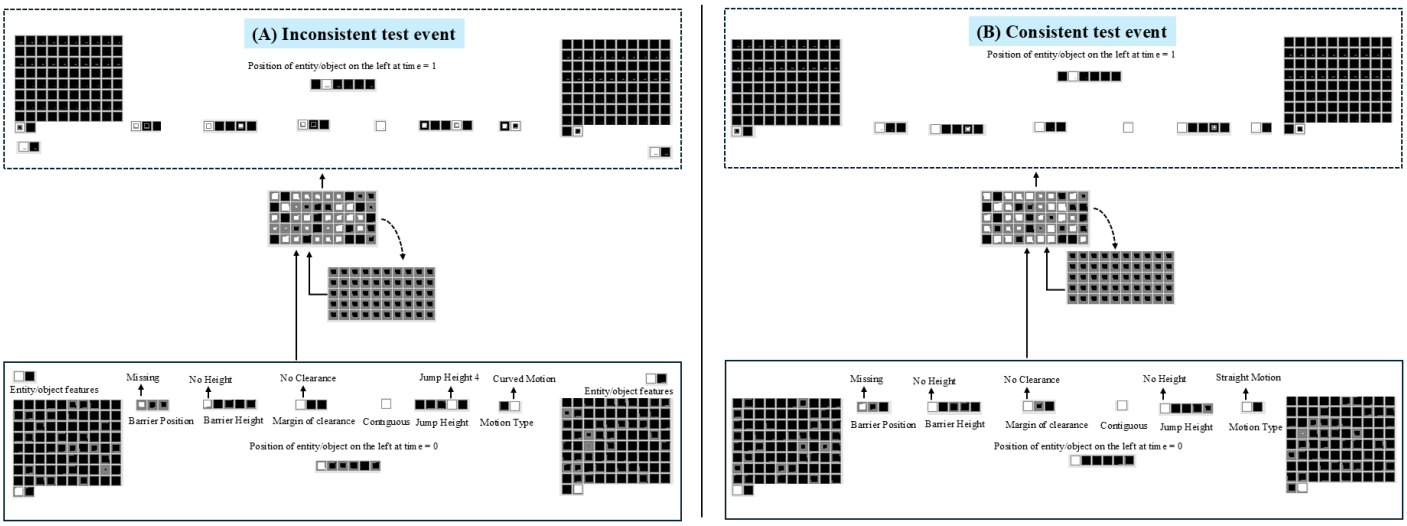

Figure 7. Depiction of the test performance of networks in the Experimental condition. (A) Network's test performance on the Inconsistent test event. (B) Network's test performance on the Consistent test events. The size of the colored region within each output unit corresponds to the activation of that output unit. Larger colored region corresponds to greater activation of that unit. Darker colored inner regions for output units correspond to activation values ≤0.5; lighter color inner regions correspond to activation values >0.4. The colored border that encircles each output unit corresponds to it "target" values. Dark borders correspond to target values of "0"; light colored borders correspond to target values of "1".

This fact helps to explain the network's responses to the barrierless test events. In the Inefficient test event, the first object again follows an arced path to reach the second object. This event was instantiated by turning on the Barrier Height 0 unit, the Curved Motion unit, the Barrier Missing unit, the single unit in the Contiguous group, the No Clearance unit, and the Jump Height 4 unit. Because five of these features appeared during the habituation phase—namely, Contiguous unit, both Animate Feature units, the Curved Motion Unit, and the Jump Height 4 unit—they will activate features they became associated with during that phase (i.e.,

Barrier Height 4, Jump Height 4), among others. For instance, presentation of the Jump Height 4 and Curved Motion input units leads to activation of the Barrier Height 4 unit. This can be seen in Figure 7. In particular, Figure 7 shows that in the Inconsistent test event, the network correctly activates the Barrier Missing, the Barrier Height 0, Contiguous, and Jump Height 4 output units. The first three are activated due to associations formed during pretraining; the last is activated due to associations formed with itself during habituation. At the same time, the network incorrectly activates the Barrier 4, Jump Height 0, and Straight Motion output units. This response is also expected: The Jump Height 0 and Straight Motion units were associated with their input counterparts during pretraining, and the Barrier Height 4 unit came to be associated with Jump Height 4, its input counterpart, Contiguous, and Animate Feature input units during habituation. Thus, three output units (i.e., Barrier 4, Jump Height 0, and Straight Motion) contribute to the error in the Inconsistent test event. This pattern of activation is precisely what the present theory predicts. Specifically, if infants, like the network, learn that a particular barrier height is associated with a specific type of motion (e.g., Curved Motion), then presenting one of these features at test (e.g., Jump Height 4) should reactivate the other (e.g., Barrier Height 4). If that associated feature is absent, as in the Inconsistent test event, then the network (and by extension, the infant) should detect an error and show heightened attention.

Now consider the Consistent test event. This event was implemented in the same manner except that the Straight Motion and Jump Height 0 input units were turned on. As can be seen in Figure 7, the network activates the same units as those in the Inconsistent test event. However, because the network correctly activates four of these units and incorrectly activates the remaining two units via the same associative processes described above, two rather than three output units contribute to this event's error. The fact that one fewer unit contributes to error in

the Consistent than in the Inconsistent event means the network looks less long at the Consistent test event than at the Inconsistent one. This provides a mechanistic explanation for infants' looking behaviors in the Experimental condition, but it does not explain infants' looking in the Control condition.

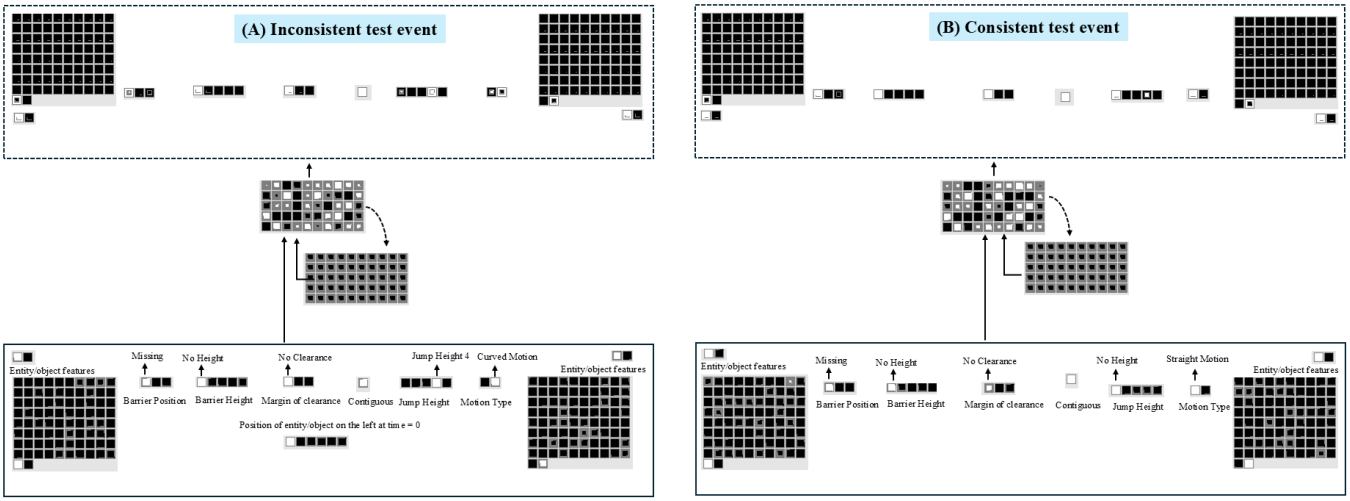

Figure 8. Depiction of the test performance of networks in the Control condition.

As it turns out, the same associative principles also explain why the network responds more similarly to the two test events in the Control condition. Figure 8 shows the test performance of one of the 20 networks in this condition. The habituation event in this condition was identical to that in the Experimental condition with one key exception: the barrier was now positioned to the side rather than in the middle. To reflect the assumption that barriers positioned to the side are less salient than centrally located ones, the learning rate for connections to and from the Barrier Height 4 units was lowered; weight decay was introduced further to decrease their salience. Consequently, the network failed sufficiently to adjust the weights connecting this unit with those features with which it was associated during habitation. A corollary of this is that

during the Inconsistent test event, the network does not activate the Barrier Height 4 or Barrier Side output units in response to input from the Curved Motion, Jump Height 4, Contiguous, and Animate Feature units, even though these features co-occurred with the Barrier Side unit during habituation. This outcome too is consistent with the present theoretical account. Because the barrier was less salient in the side position, the resulting associations with curved motion were weaker and insufficient to support reliable reactivation of the Barrier Height 4 or Barrier Side output units at test.

It might seem surprising that the network fails to activate the Barrier Side unit at all. This is because, unlike the Barrier Height 4 unit, this unit was not subject to the same salience manipulation. This failure can nevertheless be understood by considering the network's prior learning. During pretraining, the network learned that curved motion occurs over centrally located barriers. However, this regularity is violated in the habituation event for networks in the Control condition: curved motion now occurs beside the barrier. For the network to learn this new association, it would have to *unlearn* an established association between curved motion and central barriers. Ultimately, the network is unable to do this within the short time frame of the pretraining phase; that is, the short duration of the habituation phase precludes the network from overriding prior learned associations. This also explains why the network correctly activates the Barrier Missing output unit in both test events. In the Inconsistent test event, the Barrier Missing and Barrier Height 0 input units—which became associated during the pretraining phase—are presented to the model as input. Given that this activation reflects the correct response in both test events, it does not contribute to the network's error (i.e., looking at this event).

In contrast to the lowered salience of the Barrier Height 4 unit, the Jump Height 4 unit remains visually salient to reflect the assumption that infants are drawn to dynamic motion

events. As a result, the network can adjust its weights sufficiently during the habituation phase to activate the Jump Height 4 output unit in response to any of the input features it was associated with during pretraining, including itself—namely, the Curved Motion input unit, and the No Clearance input unit. If any of these units subsequently appear in a test event, the Jump Height 4 output unit will be activated. Because at least one of these input units is turned on in both test events, this output unit is active in both. This is the correct response in the Inconsistent test event but the incorrect response in the Consistent test event. Thus, the largest contributor to error in the Consistent test event is the Jump Height 4 output unit; the network correctly activates the remaining output units.

By comparison, the Inconsistent event includes two smaller contributors to error: the Straight Motion and Jump Height 0 units. These units are activated because they were previously associated with at least some of the features that appear as input to the model—namely, the Barrier Missing, Barrier Height 0, Jump Height 0, and Straight Motion input units. Importantly, the combined contribution of these two units to the network's total error in the Inconsistent event matches the contribution of the Jump Height 4 unit in the Consistent test event. As a result, the network, like infants, looks about equally at both test events. This provides a mechanistic explanation for infants' looking behaviors in the Control condition.

In sum, the reason networks in the Experimental condition looked longer at the Inconsistent test event than at the Consistent test event is that the learned association between barrier-related information (e.g., barrier height and position) and motion-related information (e.g., curved motion and high jumping) was violated in the former but not in the latter. This robust association emerged because these features were made more salient through an increased learning rate and reduced weight decay. The decision to increase the salience of these features

was itself driven by the assumption that barriers are more visually prominent when jumping actions—which are inherently attention-grabbing for infants (e.g., Slater, 1989; Vallortigara et al., 2005)—occur above them rather than beside them. By contrast, networks in the Control condition looked equally at both test events because the reduced salience of the barrier—implemented via a lower learning rate and higher weight decay—prevented the model from forming an association between the barrier and the jumping action during habituation. Taken together, these results suggest that infants' behavior in VoE Gergely-style studies may be explained by real-world experience observing curved and linear motion, combined with a domain-general associative mechanisms that operates over and forms associations among perceptual features.

Before closing, it is worth noting that the principles outlined above also account for the remaining simulations. For instance, the reason networks in the Inferring Constraints condition in Simulation 3 looked longer at the No-Obstacle test event than at the Obstacle test event was because the former violated an association that was established during the pretraining phase between jumping actions and barriers; that is, networks found anomalous test events in which jumping actions occurred in the absence of barriers because their pretraining experience was such that barriers and jumping actions went together. Similarly, the basis for networks' longer looking at the high, inefficient test event than at the low, efficient test event in Simulation 4 was that it violated a learned expectation that agents jump just high enough to clear barriers. Considered together, these findings reinforce the notion that infants' looking behaviors in VoE studies based on the Gergely paradigm may be explained by domain-general associative processes operating over perceptual regularities.

General Discussion

A question that developmental scientists have wrestled with for the last 30 years concerns how infants learn about and process others' actions. One perspective—which here I referred to as the rich perspective—maintains that infants' knowledge about agents' and objects' actions is conceptually rich and abstract and underpinned by specialized learning processes that are present from birth or shortly thereafter (Carey, 2009; Csibra & Gergely, 1998; Gergely & Csibra, 2003; Jara-Ettinger et al., 2020; Leslie, 1995; Spelke, 2022). Another perspective—referred to here as the lean perspective—argues that infants' behaviors in VoE studies, of which Gergely-style studies are an example, can be explained by domain-general learning mechanisms (e.g., associative learning, habituation, conditioning, imitation, among others) that operate over low-level perceptual features, features that are devoid of any conceptual meaning or semantic content (e.g., Benton, 2024; Benton & Lapan, 2022; Bogartz et al., 2000; Cohen et al., 2002; Cohen & Marks, 2002; Colunga & Smith, 2005; Jones & Smith, 1993; Oakes & Cohen, 2003; Perner & Ruffman, 2005; Quinn & Eimas, 1997; Rakison & Lupyan, 2008; Sloutsky, 2010).

This paper builds on this second view. The crux of my argument was that infants' performance in VoE studies that used the Gergely paradigm reflect a combination of prior experience with linear and curved motion events and online associative learning in which barriers are linked with curved action based on their salience. I argued further that the strength of these associations depends on the physical proximity of the barrier to the curved motion: Stronger associations are established when the barrier appears beneath rather than beside curved action. I then showed that when this proposal was implemented in a simple connectionist model, it was able to account for a range of important findings. Simulation 1 accounted for the finding that infants show elevated looking when animated objects behave inefficiently. Simulation 2

reproduced the finding that infants show this pattern irrespective of whether objects possess animate features. Simulation 3 captured the finding that infants can infer external constraints from an agent's actions as well as goals from its actions and external constraints. Finally, Simulation 4 replicated the finding that infants recognize that action events can differ in their relative efficiency and cost. Taken together, these simulations suggest that infants' behaviors in these studies reflect learned associations among perceptual features rather than domain-specific mechanisms that permit abstract, inferential reasoning or conceptually rich knowledge about agents, goals, efficiency, utility, or cost.

The findings presented here also contribute to a broader tradition in the developmental cognitive sciences that uses connectionist models to simulate infants' responses in VoE studies and other cognitive tasks (e.g., Benton, 2024; Benton & Lapan, 2021; Christiansen & Curtin, 1999; Cohen et al., 2002; Elman et al., 1996; Mareschal et al., 2000; Sirois & Mareschal, 2002; Schlesinger & Casey, 2003; Westermann & Mareschal, 2012; Yermolayeva & Rakison, 2014). The primary contribution of this earlier work has been to show that infants' responses in tasks ranging from object permanence to causal reasoning can be explained by associative principles. The present study extends this line of research by demonstrating that the same connectionist systems and associative learning principles can also account for infant action understanding.

Although the present model is not the first to simulate infants' behavior in Gergely-style studies, it makes several novel contributions. First, it demonstrates that a single connectionist architecture accounts for multiple findings across studies. This contrasts with the only other connectionist model in this space by Overwalle (2010); this model only simulated the Gergely et al. (1995) study. Second, by unpacking the internal dynamics of the model—specifically, how and why it generated greater error (i.e., longer looking) in response to certain events in different

conditions—it was possible to move beyond a typical demonstration that a model can reproduce behavioral data to providing a mechanistic explanation for how those data arose in the first place. A chief advantage of such an analysis is that it can provide insight into how the mind might implement associative learning. Finally, unlike earlier models that implement explicit notions of efficiency, cost, and utility, the current simulations suggest that infants' looking behaviors in action understanding studies can emerge from purely bottom-up processes. The current simulations demonstrate that these behaviors may emerge from associative learning mechanisms.

Despite the success of the current model, at least two potential criticisms are worth considering. One concerns the model's simplicity. Although it was the case that movement was represented dynamically, all other aspects of the model were represented statically and locally rather than in a distributed fashion. For example, a localist or "one-hot" encoding scheme was used to represent objects, their features, and any external constraints or barriers in the environment. These simplifications might suggest that the model is a poor approximation of infants' real-world action experience. However, a major advantage of simpler models is that it is possible to trace precisely how they process information. This was seen clearly in the section that traced how mechanistically the model reproduced infants' looking behaviors across several Gergely-style studies. That simple models can be taken apart and scrutinized in this way makes them especially well-suited for advancing theories of infant action understanding. Although it is true that complex models could better capture the richness of infants' action perception, a major disadvantage of such models is that their complexity hinders their interpretability. This complexity also makes it difficult to separate the meaningful consequences of different theoretical assumptions from extraneous model details. Simple models mostly eschew this conflict altogether by eliminating such details. Thus, although complex models may be ideal for

simulating infants' real-world action experiences, simpler models are invaluable for identifying underlying processes and advancing cognitive theory.

Another reason a simple model was chosen over a more complex one was that the larger goal of the present simulations was to shed light on how an associative learning mechanism might account for infants' looking behaviors in action understanding studies that use the Gergely paradigm. To achieve this aim, it was necessary to develop a model that could address this specific theoretical question without attempting to simulate every detail of an action event. Although a comprehensive lean account ultimately will need to explain how features such as richly structured agent representations or dynamically encoded spatial relations beyond that implemented here influence infants' performance in these studies, such features were not essential for evaluating whether sensitivity to low-level surface features and their associative relations are sufficient to explain infants' responses in these tasks. By incorporating only the most theoretically relevant features (e.g., straight and curved motion, animate and inanimate features, external constraints, etc.), the model offers a more focused test of the core claim that associative learning, combined with experience with different kinds of actions, can produce the looking patterns observed in these studies. Nonetheless, fully evaluating and extending the current theoretical account will require the development of more sophisticated models that capture the full richness and complexity of infants' real-world experiences. Additionally, although the present model is not developmental in nature, future modeling work could explore how changes in the strength of learned associations over time might account for any developmental trends observed in infants action understanding.

A second potential criticism is that because the current simulations focused on Gergely-style studies, it remains to be seen whether they can explain findings from other studies on infant

action understanding. The specific concern is that by focusing exclusively on studies based on the Gergely paradigm, the current results cannot speak to the broader literature on infant action understanding. However, this limitation may be more apparent than real. This is because if those studies can be recast within the Gergely paradigm, it is possible that the same model could account for infants' behavior in those studies as well. This would suggest that the explanatory breadth of the current extends beyond the specific paradigm for which it was originally developed. Simulation 3 took a step in this direction by reframing the Inferring Goals condition in Csibra et al. (2003)—which did not use the Gergely paradigm—within the context of the Gergely paradigm and showing that the model could nonetheless capture infants' looking behaviors in this condition. Future modeling work that builds on the current set of simulations could similarly recast other studies on infant action understanding in terms of the Gergely paradigm to determine whether the current model can explain an even wider range of findings.

More generally, the current simulations should be understood as speaking to a much broader theoretical point that is related to how infants' looking behaviors in action understanding tasks should be interpreted. The central claim is that such behaviors reflect learned associations among semantically neutral perceptual features rather than abstract inferences about agent action, goal, utility, cost, and efficiency, inferences that may be governed by innately endowed specialized processes. One implication of this view is that the current theoretical account should be implementable in any generic connectionist system, and thereby able to test whether a wider range of action understanding studies can be captured. Such a model could offer a perceptually grounded, theoretically coherent alternative to accounts that rely on domain-specific knowledge to explain infants' responses across paradigms.

In sum, the current studies were carried out to test a novel theoretical account of infants' action understanding that is grounded in domain-general associative learning and real-world experience with different types of actions and that builds on and extends existing lean accounts in the literature. By successfully modeling infants' responses in studies using the Gergely paradigm, this work challenges the notion that infants rely on conceptually rich knowledge and specialized systems for processing action. Instead, the findings suggest that associative learning is sufficient to explain infants' responses in these studies. It may also offer a more parsimonious explanation of infants' behavior in these studies than accounts that invoke abstract notions of cost, efficiency, and agent action. More broadly, this work speaks to ongoing debates about how best to interpret infants' looking behaviors in VoE tasks, perhaps as well as in other cognitive tasks. If such behaviors can be captured by simple associative processes that operate over perceptual input, this casts doubts on claims that infants' looking in these tasks necessarily reflect conceptually rich inferences. But the debate continues.

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